Day 1

NCHC X OpenACC X NVIDIA

OpenACC
More Science, Less Programming

2026 NCHC Open Hackathon

Important Dates

- ~~10 / 14 (Wed) : Day 0: Kick-off Meeting 15:00~17:00 PM (Online)~~
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  - 15:00 - 15:10 : All-hand meeting
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NVIDIA GTC Berlin 2026

OCT 20 -22

https://www.nvidia.com/en-eu/gtc/keynote/

Keynote at 5-7 PM OCT 21, Taiwan Time Zone.

NVIDIA is bringing the world's most essential AI developer conference to the heart of Europe. Conference programming will span the full AI stack, from infrastructure to the industries being transformed by AI.

CEO Jensen Huang's keynote at the iconic Tempodrom will set the stage as thousands of developers, researchers, startups, and industry leaders convene at STATION Berlin for sessions, labs, and networking.

Don't miss your opportunity to be part of GTC Berlin, where the global ecosystem from every layer of the AI stack come together to shape what's next.







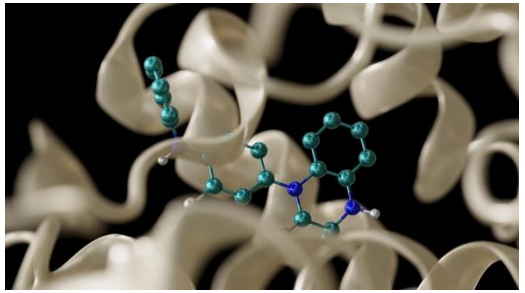


OCT 20 -22 GTC Berlin Key Content Topics



Agentic AI and Claws

Learn from industry experts how to build specialized AI agents that reason, plan, and act to deliver real business results with Claws, open models, libraries, frameworks, and platforms.



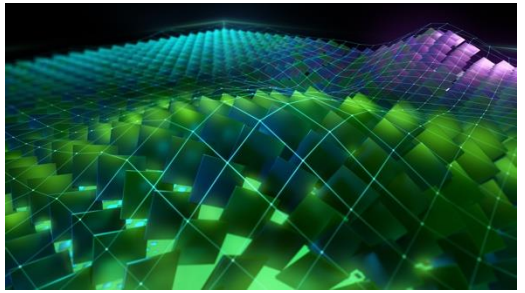
AI for Science

Discover how accelerated computing and AI are enabling the AI scientist. You'll learn how they fuse agentic systems, simulation, and data to unlock breakthroughs in climate, materials, biology, and beyond.



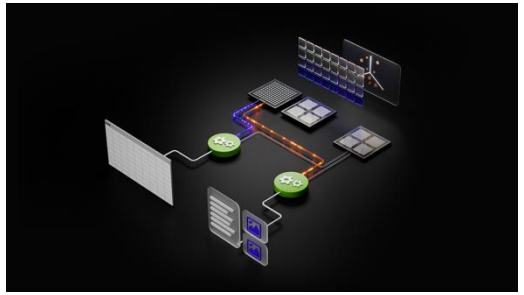
AI Infrastructure

See how next-generation infrastructure can turn your data centers into scalable intelligence engines that efficiently and securely produce, manage, and deploy AI across your enterprise.



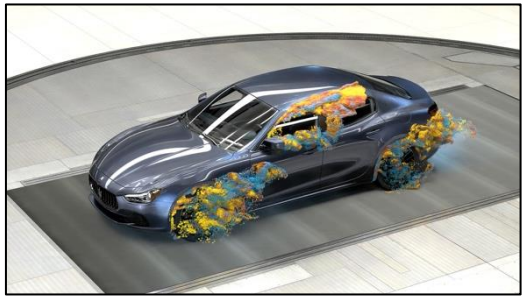
CUDA and Libraries

Dive into NVIDIA CUDA™, open source frameworks, libraries, and developer tools that let you build, optimize, and deploy GPU-accelerated applications, from community-driven projects to the latest open models and SDKs.



Data Processing

Explore how NVIDIA cuDF and NVIDIA cuVS accelerate agentic AI, processing terabytes of structured and unstructured data faster on the platforms you already use.



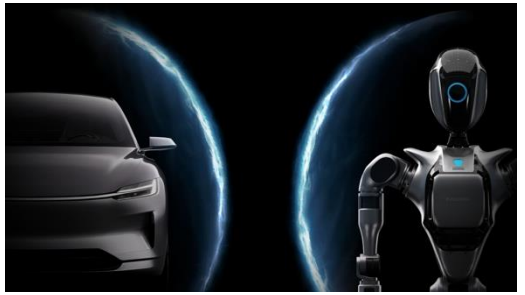
Industrial AI

See how NVIDIA accelerated computing, CUDA-X libraries, AI models, and digital twins help engineers design, simulate, and optimize products, factories, and infrastructure.



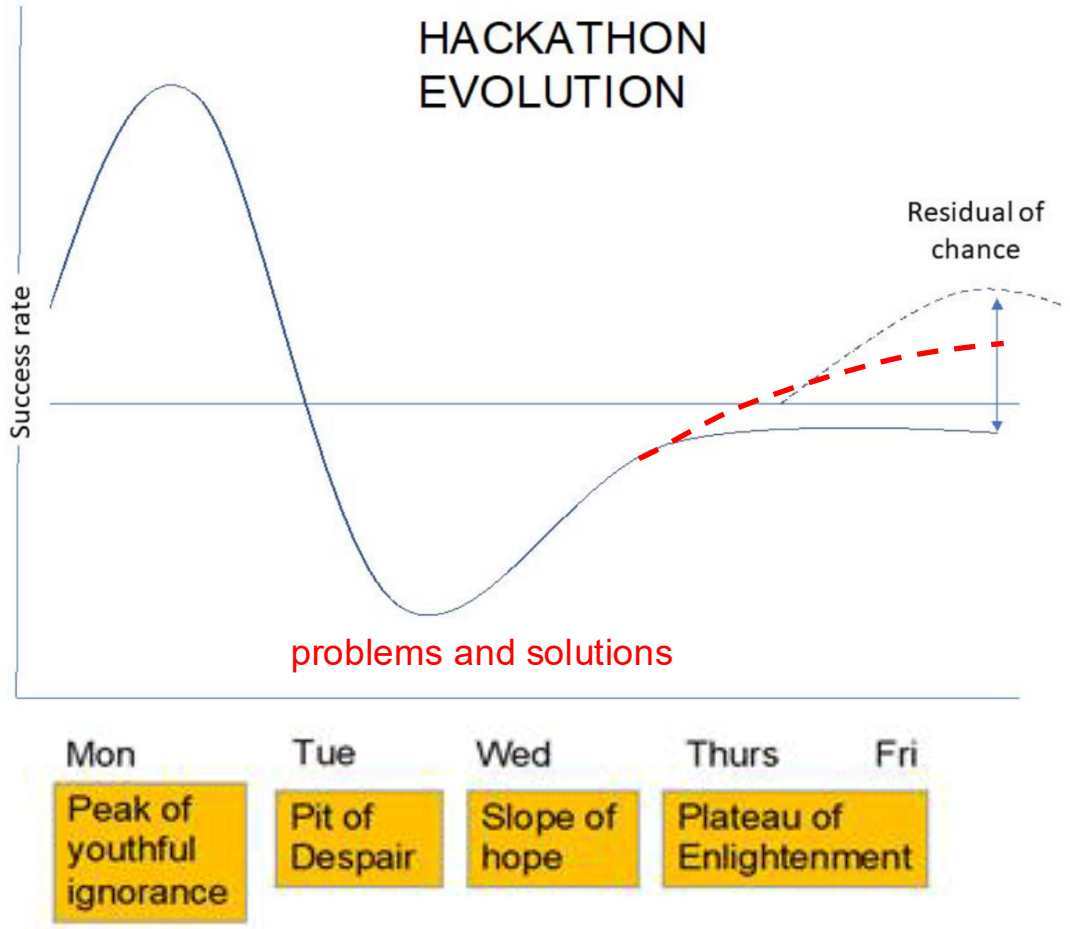
Open Models and Datasets

Explore the new universe of open models, data, and tools that let you customize deeply, integrate quickly into your agent architectures, and deploy trustworthy AI tailored to your domain.



Physical AI and Robotics

Discover how quantum computing and HPC, unified on accelerated platforms, enable hybrid workflows that push the frontiers of science, engineering, and industry at unprecedented scale.



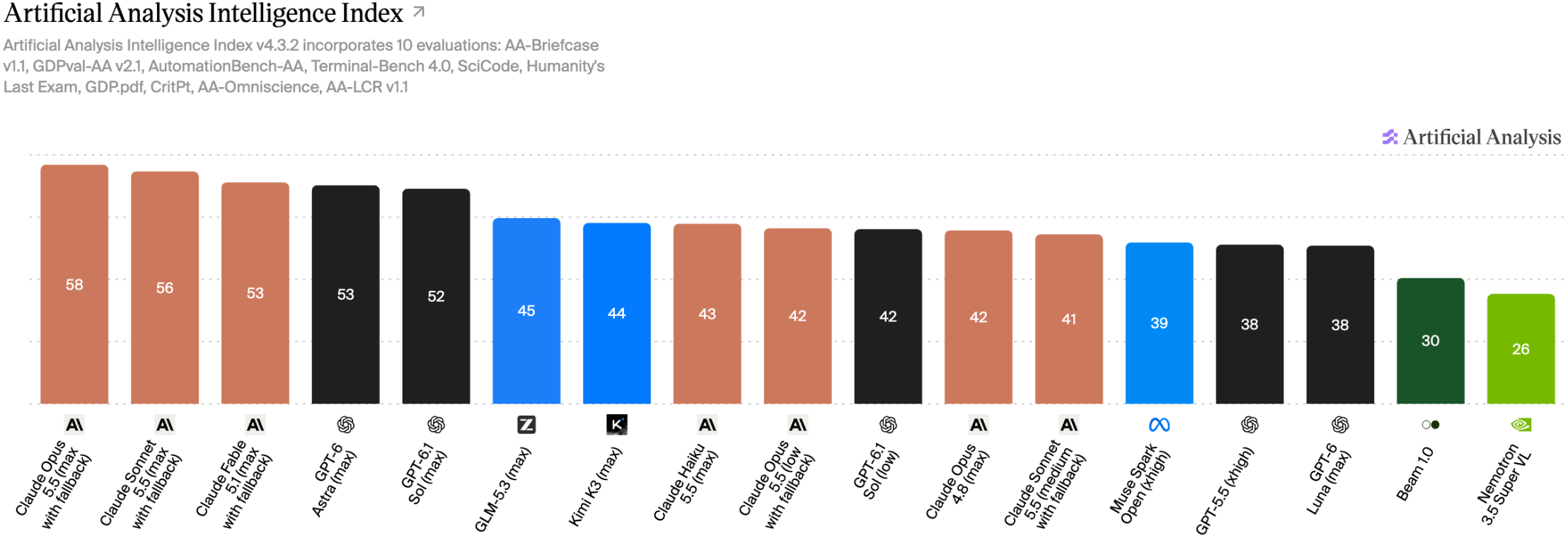
This article highlights the Oak Ridge Leadership Compute Facilitys GPU Hackathon
https://www.computer.org/csdl/magazine/cs/2018/04/mcs2018040095/13rRUxYIN88

Overview to Computing Resources

- NCHC Nano4 for H200 GPU, Forerunner-1 for Grace CPU
  - NT$ 5,000 for each iService account by default.
  - **Request more by demand**. (no more than NT$ 20k)
- NCHC RAP LLM API
  - NT$ 20,000 for each Team by default.
  - **Request more by demand.** (no more than NT$ 40k)
- User Guide and limitations
  - H200 GPU Nano 4: https://man.twcc.ai/@nano4-manual/SJM_FuxDWe
  - Grace CPU Forerunner-1: https://man.twcc.ai/@f1-manual/partition
  - Check your wallet status: https://iservice.nchc.org.tw/nchc_service/nchc_service_qa_single.php?qa_code=381
- Key Reminders
  - Run agent on compute node,
    DO NOT run agent or service on login node!!
  - Release resource if you didn't use it to save credits.

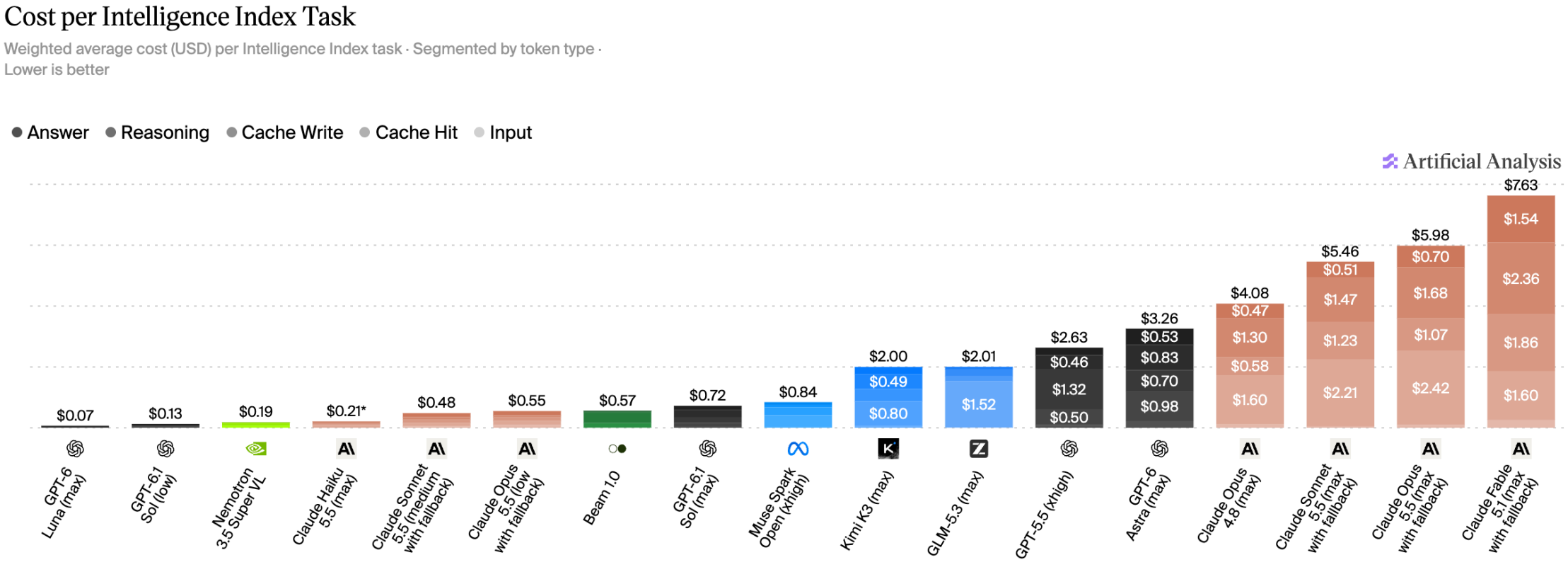
Independent Analysis of AI - Index

https://artificialanalysis.ai/



Independent Analysis of AI – Cost per task

https://artificialanalysis.ai/



OpenACC More Science, Less Programming
OPEN HACKATHONS
NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing
NVIDIA

Nsight System: AI Assisted Analysis (preview feature toy demo)

https://docs.nvidia.com/nsight-systems/AnalysisGuide/index.html#ai-assisted-analysis

範例程式核心：每次 launch 後都等待

```
__global__ void increment(int* values, int count) {
  int i = blockIdx.x * blockDim.x + threadIdx.x;
  if (i < count) values[i] += 1;
}
for (int i = 0; i < 1000; ++i) {
  increment<<<1024, 256>>>(device_values, 262144);
  cudaDeviceSynchronize();

}
```

省略配置、暖機、錯誤與輸出檢查；完整程式以 cudaProfilerStart/Stop 標記量測區間。

AI-assisted analysis 在本例提供的幫助

讀資料 整理 API / kernel 次數與耗時，附 report 證據。

找模式 辨識 **1,000 次同步等待**，對應左上程式。

提建議 同一 stream、CPU 不需中途讀回時，
將同步移到迴圈外，再重新量測驗證。

01 擷取暖機後的 CUDA / NVTX report

```
nsys profile --trace=cuda,nvtx \
  --sample=none --cpuctxsw=none \
  --discard-environment=true \
  --capture-range=cudaProfilerApi \
  --capture-range-end=stop \
  -o reports/before ./toy_sync before
```

02 nsys --help 取得 skill 路徑，再提問

```
請依照 [貼上 SKILL.md 完整路徑] 分析
reports/before.nsys-rep：
列出 API 與 kernel 次數、耗時及證據，
提出最小修改與驗證方法。此輪只分析。
```

自己的 AI agent 讀取 NVIDIA 隨附的
`skills/nsight-systems/SKILL.md`，依官方流程分析。

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https://docs.nvidia.com/nsight-systems/AnalysisGuide/index.html#ai-assisted-analysis

BEFORE

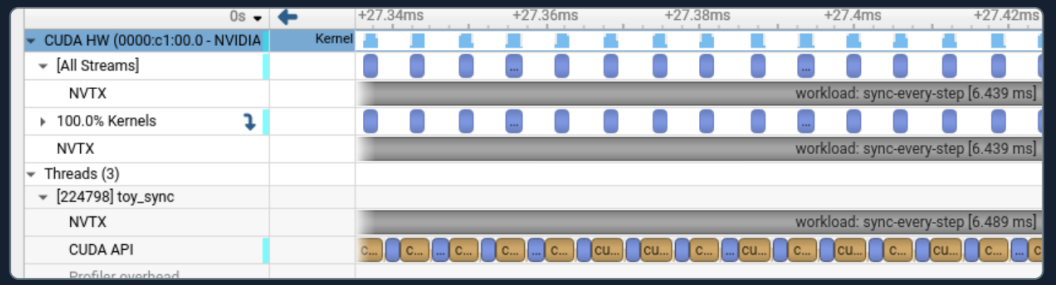
每一步都同步

5.037 ms

未開 profiler · workload 中位耗時

`kernel → sync → kernel → sync → …`

Nsight Systems 實機截圖 · 局部時間軸



同步次數	1,000	Kernel 次數	1,000	Kernel 總時間	1.638 ms

程式片段（簡化示意）

```
for (...) {
  increment<<<...>>>(values);
  cudaDeviceSynchronize();
}
```

AFTER

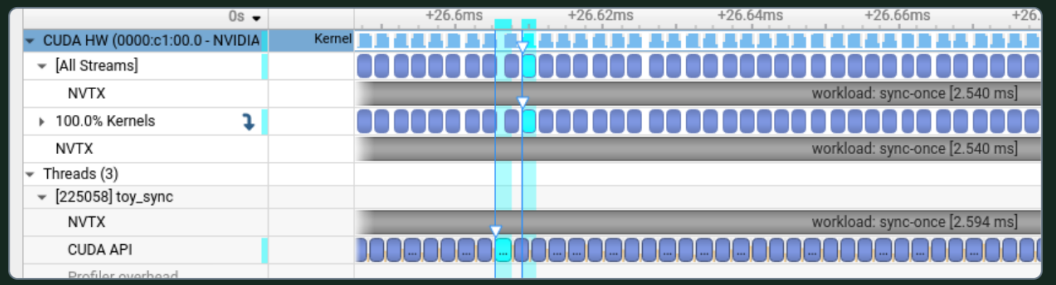
全部提交後同步一次

1.992 ms

未開 profiler · workload 中位耗時

`kernel → kernel → … → 最後 sync`

Nsight Systems 實機截圖 · 局部時間軸



同步次數	1	Kernel 次數	1,000	Kernel 總時間	1.634 ms

程式片段（簡化示意）

```
for (...) {
  increment<<<...>>>(values);
}
cudaDeviceSynchronize();
```

2.53× **暖機後 workload 加速**

兩個版本的 262,144 個輸出元素均檢查通過；kernel 工作量與總耗時相近。

OpenACC More Science, Less Programming · OPEN HACKATHONS · NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing · NVIDIA

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Nov 12 Final Day Lunchbox and Attendance

2026 NCHC Open Hackathon - Lunchbox Survey – Fill out form

All attendees are required, including mentors, please make sure reply it by this Fri, Oct 23, thanks. T

